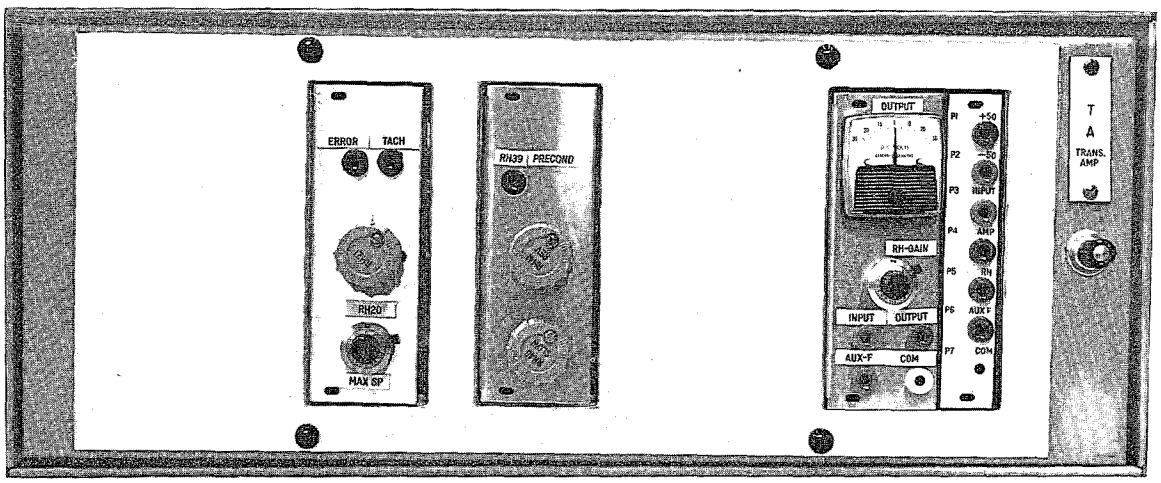
INSTRUCTIONS

GEK-7742

TA TRANSISTOR PREAMPLIFIER



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GENERAL ELECTRIC

CONTENTS

	Page
INTRODUCTION	3
DESCRIPTION	3
OPERATION	6
Input Card (IC3600SQIC1)	7
Operational Amplifier Card (IC3600AOAC1)	7
Antihunt Card (IC3600SAHD1)	8
Auxiliary Function Card (IC3600SQIA1)	8
Meter and Test Features	10
INSTALLATION	11
INSTALLATION ADJUSTMENTS	11
Maximum Speed Adjustment	11
Precondition Adjustment	12
Time Constant Adjustment	12
Gain Adjustment	13
Acceleration Adjustment	13
Slow and Reverse Adjustment	13
Slack Takeup Adjustment	14
Preamplifier Zero Adjustment	14
Slowdown Clamp Removal	14
Special Features	14
Special Connections	14
TROUBLESHOOTING	15
General	15
Procedure	16
APPENDIX A	
Auxiliary Function Amplifier Card (IC3600SQIA1)	19
APPENDIX B	
Operational Amplifier Card (IC3600AOAC1)	21
APPENDIX C	
Antihunt Card (IC3600SAHD1)	26
APPENDIX D	
Input Comparison Card (IC3600SQIC1)	27
APPENDIX E	
Jumper Card (194A3396G1)	28
APPENDIX F	
Test Card (194A3397G1)	28

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

TA TRANSISTOR PREAMPLIFIER

Before any adjustments, servicing, parts replacement or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, the POWER SUPPLY MUST BE DISCONNECTED.

INTRODUCTION

The TA transistor preamplifier is designed to amplify an extremely small error voltage at a sufficient level to excite a power amplifier in a paper mill drive. This preamplifier is one unit of a high-accuracy speed regulator where a tachometer voltage is compared to a master speed-reference voltage. The difference between these two voltages is amplified to produce a speed error voltage. A comparison circuit that detects the voltage difference is located within the TA. Also contained within the preamplifier are components to amplify the error voltage with sufficient gain to stabilize the regulator.

In addition to the amplification function, the preamplifier provides the following control functions:

1. Precondition Function. These circuits are used when the preamplifier is not required to regulate. In this mode of operation, the preamplifier is ready to accept a command to run, slow, or reverse.

2. Acceleration Function. During a transition from preconditioning to run, or slow to run, limitations are placed on the rate of change of the tachometer input voltage. This rate of change limitation is controlled by a fixed RC circuit in the preamplifier.

3. Slow, Reverse and Slack Takeup Functions. The preamplifier contains those components which perform the slow and reverse speed reference functions as well as the slack takeup. These circuits are controlled by external electrical interlocks.

TEST CIRCUITS

The TA has a test card which is removed from its storage receptacle whenever tests are performed. A jumper card is removed and the test card inserted in this slot. External test equipment can then be used at the test points mounted on the test card. These equipment tests are performed on a go/no-go basis.

DESCRIPTION

All components of the transistor preamplifier, except the remotely located electrical interlocks and slack takeup push button, are mounted inside an 8-inch high by 20-inch wide steel enclosure, commonly called a module. This module is designed to mount on control panels where needed in a paper mill drive. Four mounting bolts which provide for convenient removal and replacement secure the module to the panel. All wires entering and leaving the module are connected to screw-type terminals accessible from the front when the cover plate is removed.

Those components visible from the front of the preamplifier under operating conditions are shown in Fig. 1. When the front cover plate is removed,

PREAMPLIFIER SPECIFICATIONS

Power Input	+50V, 36 ma + output current -50V, 53 ma + 5.5 ma in slack takeup
Reference Input	-3 to -30V, 0 to -200 microamperes in run mode - except coater operation (see Special Connections - Installation Adjustments) 0 to -30V, +70 to -440 microamperes in precondition
Regulated Voltage Input	50 to 350 volts maximum
Drift (Regulated Voltage)	.01 percent/°C on regulated voltage
Lead Time Constant	0.165 to 6.25 sec. adjustable
Ripple (Maximum Tolerable on) (Regulated Voltage)	10 Volts p-p @ 30 cps; 200V tachometer V_{pp} = 0.00179 x frequency in cps x reg. rated volt
Load (Operational Amplifier Output)	±30V, 10 ma maximum
Precondition Output	-0.5V to +0.5V adjustable
Acceleration	2 to 32 volts/sec adjustable, regulated voltage
Slow Adjustment	+1 to +14 volts adjustable, regulated voltage
Reverse Adjustment	-0.3 to -17 volts adjustable, regulated voltage, depending on slow adjustment setting (0.3 to 1.2 times slow voltage)
Slack Takeup Adjust	0 to 3 percent adjustable, regulated voltage; polarity depends on input voltage

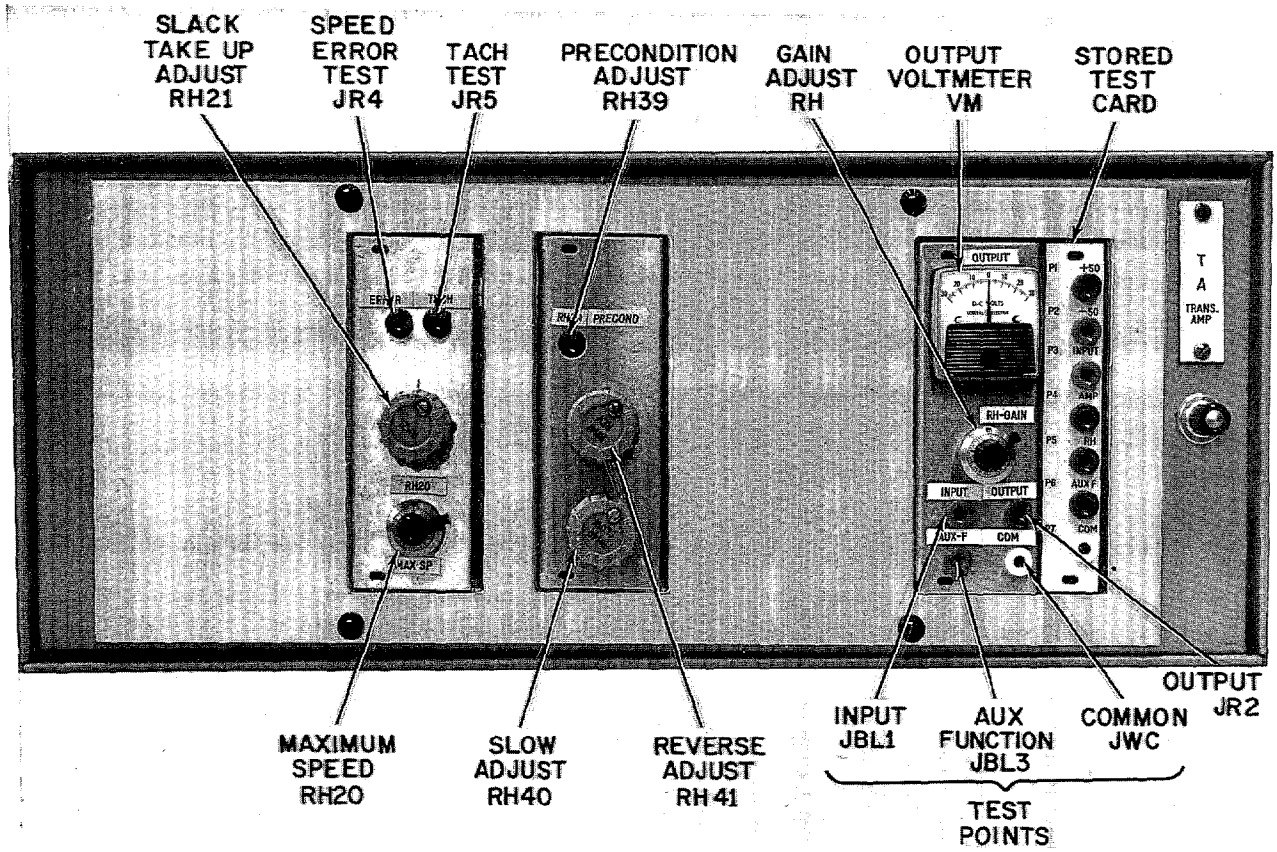


Fig. 1. Front view of TA transistor preamplifier

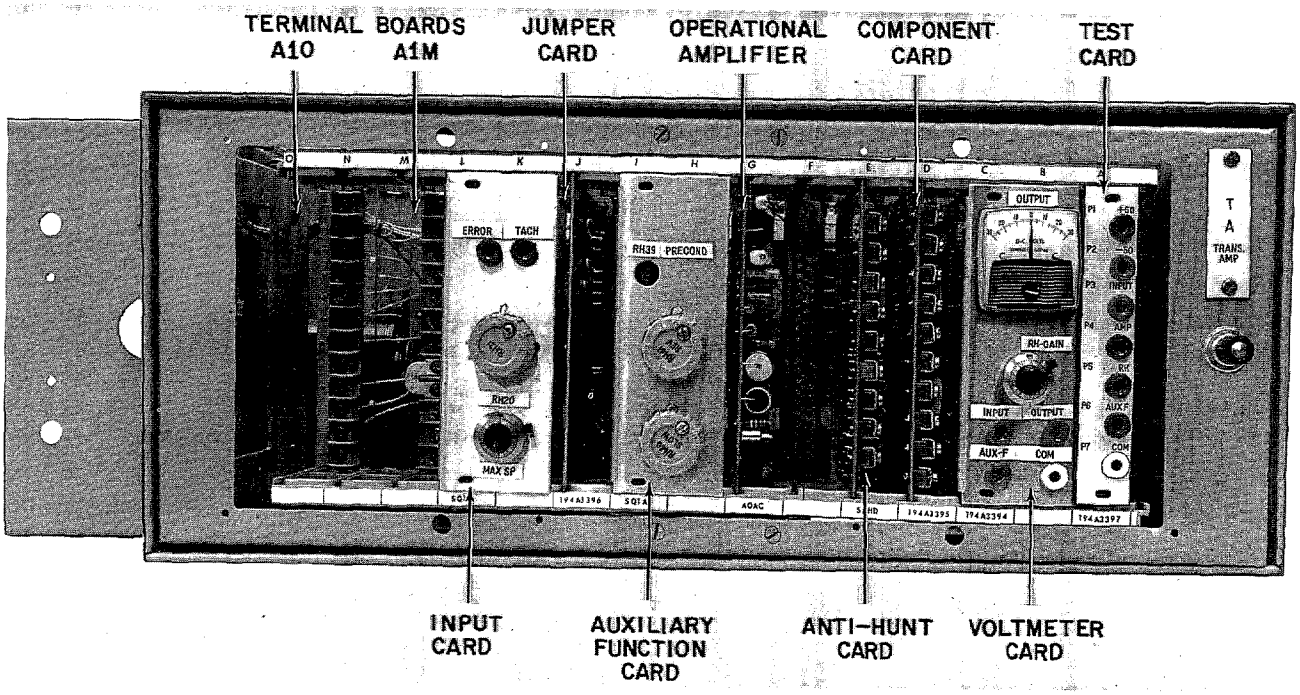


Fig. 2. Front view of TA transistor preamplifier with cover plate removed

front access is obtained to the terminal boards and printed-wiring boards. These components are shown in Fig. 2 and identified in Table 1.

A hinged door with latch provides access to all wiring and devices inside the enclosure as shown in Fig. 3. The printed-wiring board pins plug into

TABLE 1

DEVICE LOCATION AND IDENTIFICATION

SLOT DESIGNATION	DEVICE NAME	IDENTIFICATION NUMBER
A1O	Terminal board A1O	None
A1N	Not used	-
A1M	Terminal board A1M	None
A1L	Input card	IC3600SQIC1
A1K	Not used	-
A1J	Jumper card	194A3396G1
A1I	Auxiliary function card	IC3600SQIA1
A1H	Not used	-
A1G	Operational amplifier	IC3600AOAC1
A1F	Not used	-
A1E	Anti-hunt card	IC3600SAHD1
A1D	Component card	194A3395G1
A1C	Voltmeter card	194A3394G1
A1B	Not used	-
A1A	Test card	194A3397G1

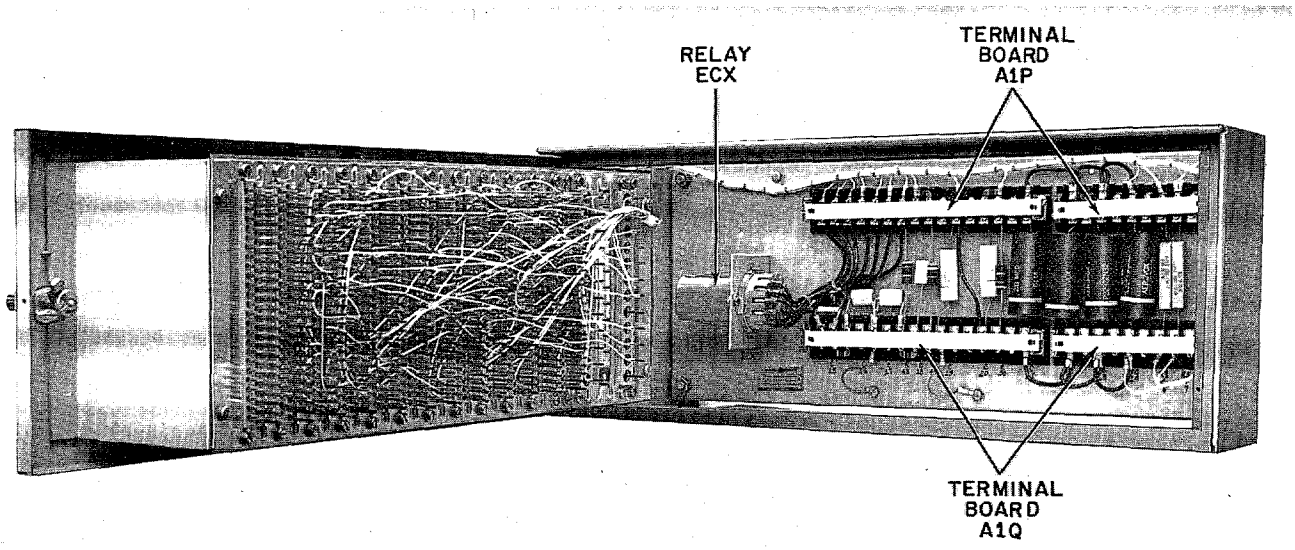


Fig. 3. TA transistor preamplifier with door open showing internally mounted devices

a matching receptacle mounted at the rear of the door assembly. All interconnecting wires between printed board receptacle pins are installed by means of a gas-tight, pressure connection utilizing a special pressure gun as a tool. This tool simultaneously strips the insulation from the conductor, attaches the lug, and compresses the lug and wire to the pin with a wipe-clean action. The conductor is held, under high pressure, against the surface of the connector pin to provide a gas-tight connection between conductor and pin.

The printed-wiring boards are made of glass epoxy to resist moisture absorption and warping in mill environments. After the board components are assembled and soldered, the entire board is conformal coated to protect it against dirt, moisture, fumes and other environmental contamination. When the boards are removed, a special card puller supplied with the equipment releases the printed board pins from the receptacle. During replacement, the back of this puller helps to seat the board pins into the mating receptacle. Both the board and receptacle contact pins are nickel-tin plated for good electrical contact and corrosion resistance.

OPERATION

As shown in Fig. 4, the TA transistor preamplifier contains four basic circuits mounted on plug-in cards:

1. Input
2. Operational amplifier
3. Antihunt
4. Auxiliary function

In addition to these basic cards, the preamplifier contains four secondary cards as follows:

1. Voltmeter card
2. Component card
3. Test card
4. Jumper card

All of the components shown in Fig. 4 within the double dashed line are located inside the preamplifier enclosure. The externally-mounted devices include the electrical interlocks and the slack take-up push button. Power is supplied to the preamplifier from an external ±50-volt power supply.

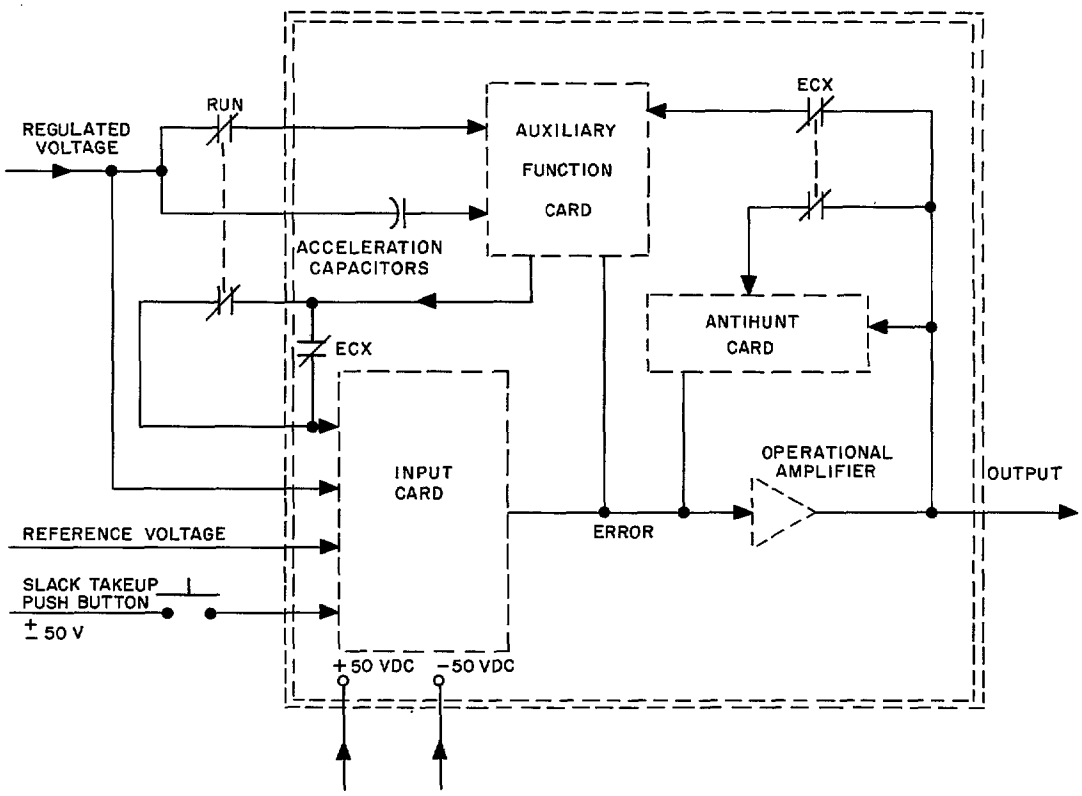


Fig. 4. TA transistor preamplifier, simplified diagram

INPUT CARD (IC3600SQIC1)

The comparison circuit on the input card is a resistor bridge as shown in Fig. 5.

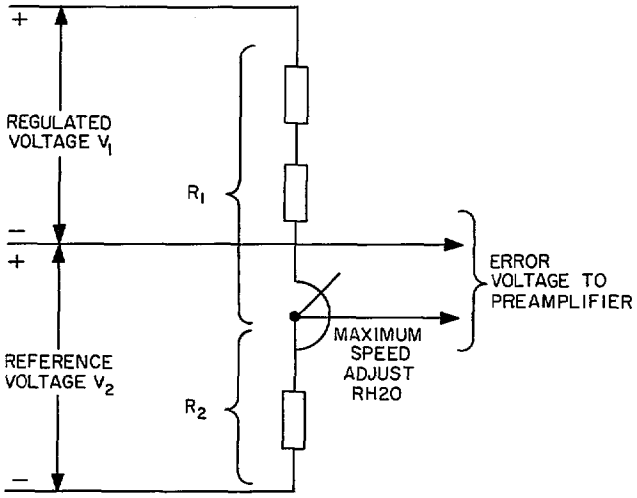


Fig. 5. Input comparison circuit

When the regulator is operating correctly, the error voltage will be essentially zero for steady-state operation and no current is received at the amplifier from the slider of the maximum-speed-adjust potentiometer. Since the slider is set for zero voltage and no current flows from the slider, the current through R_1 will equal the current in R_2 as follows:

$$\frac{V_1}{R_1} = \frac{V_2}{R_2} \text{ or } \frac{V_1}{V_2} = \frac{R_1}{R_2}$$

For a given value of fixed resistors in the bridge, rotating the maximum-speed-adjust potentiometer causes R_1 to increase resistance. Therefore, the regulated voltage V_1 must increase for a fixed reference voltage V_2. The fixed portion of R_1 is adjustable by means of jumpers on the input card to provide for different maximum regulated voltages.

The slack takeup adjustment is also located on the input card as shown in Fig. 6. When the slack takeup push button is depressed, a voltage is applied to the slack takeup potentiometer. The setting of this potentiometer establishes the current flow through RA and RB. If a negative 50 volts is applied through the push button, this current will cause the regulated voltage to increase.

A time delayed slack takeup is sometimes required when a non-reversing Silcomatic drive is used. In this application, the tension current in the motor is not sufficient to slow down the drive without reducing the current to zero. If this time constant is desired, capacitor C is connected into the

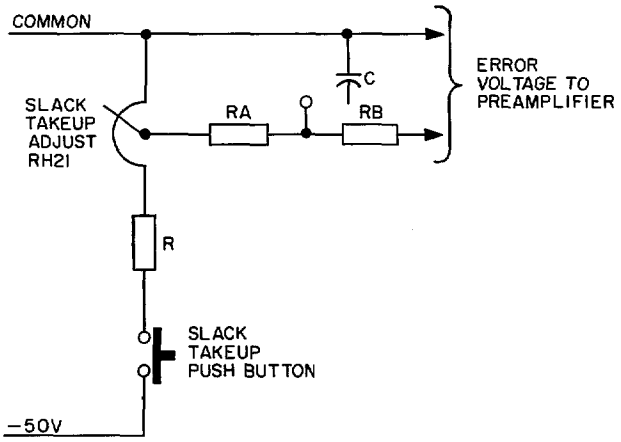


Fig. 6. Slack takeup circuit on input card

circuit to supply the time delay. See slack takeup adjustment under installation instructions.

OPERATIONAL PREAMPLIFIER CARD (IC3600AOAC1)

The operational amplifier is a five-stage, transistorized, high-gain, inverting d-c amplifier. Both the amplifier symbol and basic circuits are shown in Fig. 7. This card amplifies the voltage, V_G, which is impressed on the input from card pin 16 to common. The amplifier produces an output voltage, V_O, with a polarity at pin 46 opposite that of the polarity at pin 16. For a swing in input voltage of ± 0.8 millivolt, the output voltage will swing ± 30 volts. With this high gain, the input voltage to the preamplifier can be assumed to be essentially zero for purposes of description of operation.

When feedback is used around the amplifier, the gain of the amplifier is modified by the impedance of the feedback network as shown in Fig. 8. Since the amplifier input, V_G, is approximately zero and the input impedance to the amplifier is high, i_e is approximately zero; therefore, $i_O = i_E$.

$$\text{Also, } i_O = \frac{V_O}{R_O} \text{ and } i_E = \frac{V_E}{R_E}$$

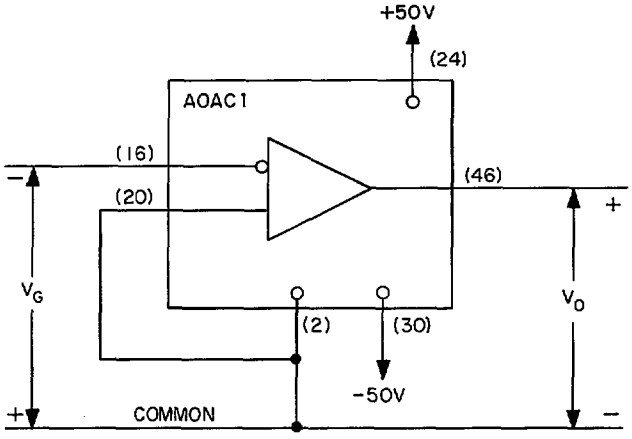


Fig. 7. Operational preamplifier with printed board connections

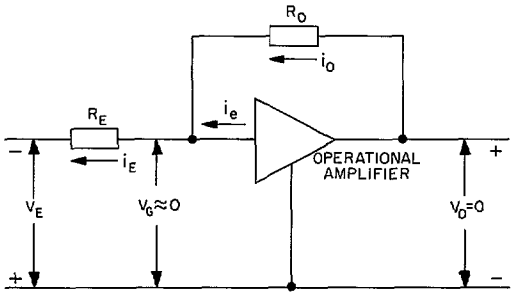


Fig. 8. Operational amplifier showing input and feedback resistors

$$\text{Therefore, } \frac{V_O}{R_O} = \frac{V_E}{R_E}$$

$$\text{or} \qquad V_O = (R_O) \times \frac{(V_E)}{(R_E)}$$

The operational amplifier therefore amplifies the input current, $\frac{V_E}{R_E}$ by the quantity R_O to produce an output voltage V_O. This output voltage V_O has the opposite polarity of V_E. A combination of resistors and capacitors in the feedback circuit produces an output gain which varies with frequency. This feedback network is used to stabilize the regulator.

Additional information including the elementary diagram, circuit characteristics and application notes are contained in the Appendix of these instructions.

ANTIHUNT CARD (IC3600SAHD1)

The antihunt card contains the resistor and capacitors in the feedback circuit of the operational amplifier. The amount of feedback is controlled by the gain-adjust potentiometer, RH, located on the voltmeter card. This basic circuit is shown in Fig. 9 and it is this configuration of feedback components that produces the stabilization required for the regulator. With d-c input error voltages, V_E, the full gain of the amplifier is used. In this application, the capacitor blocks d-c flow except for leakage resistance and there is no feedback. For a-c error voltages, the feedback components supply the time constants necessary to stabilize the regulator.

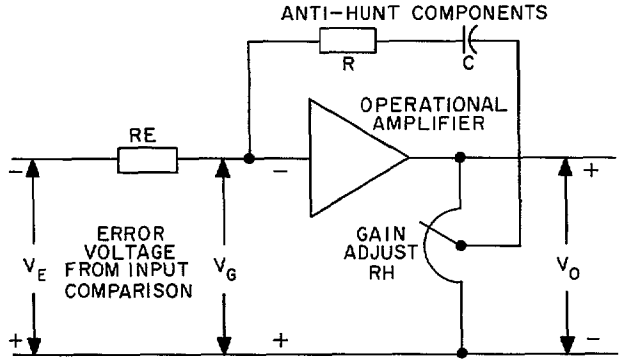


Fig. 9. Operational amplifier showing anti-hunt components and gain control

AUXILIARY FUNCTION CARD (IC3600SQIA1)

The auxiliary function card has as its major component a unity voltage gain amplifier. In combination with this amplifier are several configurations of resistors and capacitors used for setting slow speed, both forward and reverse, and section acceleration. There is also a 10 to 1 attenuation bridge on the output of the unity voltage gain amplifier. A silicon diode is used to remove the auxiliary function output from the preamplifier after it has served its purpose. Those auxiliary functions that are performed and the methods used to accomplish these functions in a speed regulating system are described in the following paragraphs.

The TA transistor preamplifier is used for several auxiliary functions. These functions are preconditioning, timed acceleration, slow-speed regulation and reverse-speed regulation. All of these auxiliary functions are accomplished through the unity voltage gain amplifier, attenuator, and diode connected to the point where the input comparison error voltage is impressed on the operational amplifier. These auxiliary function circuits are so interlocked with the magnetic control that the auxiliary function amplifier will override the input comparison error voltage when the auxiliary functions are required.

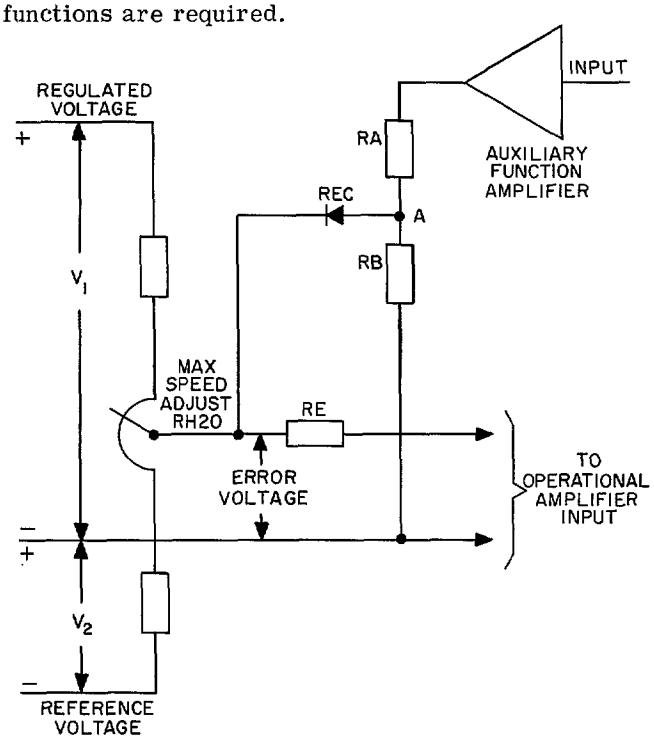


Fig. 10. Auxiliary function amplifier connected to input comparison circuit

Fig. 10 shows the auxiliary function amplifier connected into the input comparison circuit. When the voltage at point A is less positive than the error voltage, the diode, REC, will block and the unity voltage gain amplifier will not control the error voltage. Under this condition, the reference voltage has full control of the error voltage.

Preconditioning

The TA is designed to regulate its output voltage level between zero and +0.5 volt whenever an error signal is present from the input comparison bridge. Preconditioning prepares the preamplifier for smooth control when it is released for regulation in the over-all system.

To accomplish this preconditioning, the output signal of the operational amplifier is fed back into the auxiliary function amplifier so that it can control the current through diode, REC. The reference side of the input comparison bridge is clamped to a fixed voltage. The preconditioning circuit is shown in Fig. 11.

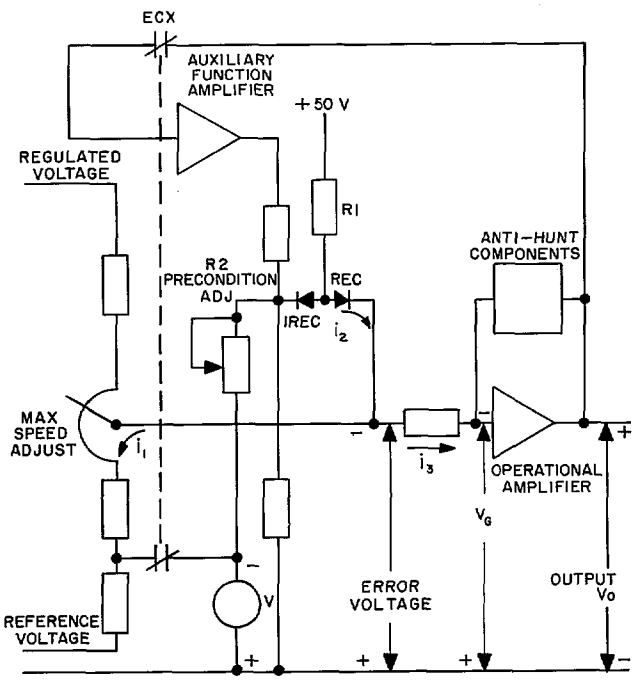


Fig. 11. Preconditioning circuits

The resistor R1 supplies current that passes through both diodes, 1REC and REC. When the current i_2 through REC equals the current i_1 established by the bias voltage V, the current i_3 into the operational amplifier is zero. Since voltage V_G is very nearly zero, the error voltage will also be zero when i_3 equals zero. The precondition adjust potentiometer, RH39, is adjusted to turn on 1REC to allow the output of the operational amplifier to regulate i_3 to zero. This precondition adjust potentiometer is set to maintain from 0 to +0.5 volt output range of the operational amplifier.

Timed Acceleration

When the precondition circuit is released, interlock ECX opens and releases the operational amplifier for regulation in the over-all system.

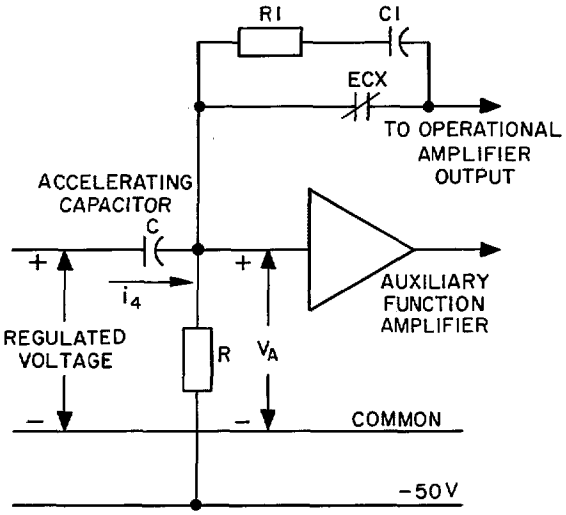


Fig. 12. Acceleration input circuit to auxiliary function amplifier

Fig. 12 shows the acceleration circuit input to the auxiliary function amplifier.

After interlock ECX opens, the input comparison bridge is released to the reference voltage (see Fig. 10). Also, the input voltage V_A (Fig. 12) to the auxiliary function amplifier begins to drop to the -50V bus level at a time constant determined by R and C. As V_A drops, the current through 1REC (see Fig. 11) starts to increase, thus decreasing the current i_2, through diode, REC. When i_2 decreases, the input current, i_3, to the operational amplifier tends to increase the regulated voltage. This change in regulated voltage produces a current through the accelerating capacitor, C. The current through the accelerating capacitor tends to increase V_A to maintain the proper current, i_4, to accelerate the drive at a constant rate of change of regulated voltage. The rate is adjustable by changing the value of the accelerating capacitor, C. Capacitor C1 prevents the input to the auxiliary function amplifier from changing abruptly.

When the regulated voltage has accelerated to the level required to produce zero error voltage from the input comparison bridge, the current i_2 through diode REC (Fig. 11) is reduced to zero. The auxiliary function amplifier no longer has control of the error voltage. Simultaneously, the input voltage, V_A, to the auxiliary function amplifier drops toward the negative bus (-50V) and sustains a reverse voltage on REC.

Slow and Reverse

In the slow, or reverse mode, the bias voltage on the reference side of the input comparison

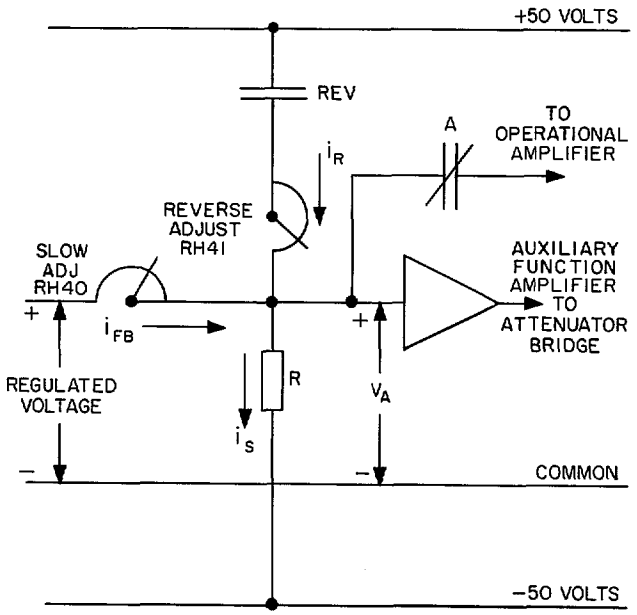


Fig. 13. Slow and reverse circuits

bridge is not released. The input to the auxiliary function amplifier becomes the comparison point for the speed regulator as shown in Fig. 13.

Input voltage, V_A, the regular error voltage to the auxiliary function amplifier functions as a second error input voltage with V_A varying plus or minus one volt. In the slow mode, a fixed reference current, i_S, is produced by resistor R while the regulator holds i_{FB} in the direction shown through the slow-adjust potentiometer. Increasing the resistance of the potentiometer will cause the tachometer voltage to increase in order to maintain the proper level of i_{FB}.

Once the slow adjust setting is established, energizing the reverse relay will supply an additional current, i_R, into the auxiliary function amplifier tending to raise the input voltage, V_A. This increased input causes the operational amplifier to produce a negative output and the motor rotates in the opposite direction. The direction of i_{FB} is thus reversed so that $i_R - i_S = -i_{FB}$. When the reverse-adjust potentiometer is varied, the reverse voltage is established.

On non-regenerating Silcomatic sectional drives, reverse operation is normally accomplished by reversing the tachometer voltage with relay interlocks and using the slow circuit to establish reverse speed. For this type of operation, the reverse input is not used.

Slowdown Clamp

In a non-regenerating Silcomatic sectional drive, the mill process may require that the current in the motor goes to zero. When this occurs, the TA loses control of the section motor speed. If the mill process maintains the motor current at zero, the output of the TA will go to negative saturation (greater than -30 volts) in its attempt to slow down the drive. When the motor current returns, high current transients will result in the attempt to regain speed control. To prevent these transients, a slowdown clamp is provided on the TA transistor preamplifier. This clamp limits the output of the TA to approximately 2.5 volts below the level required to produce current in the motor; thus preventing the output from going to negative saturation.

The slowdown clamp circuit is shown in Fig. 14. When the Silcomatic drive is producing motor current, the motor terminal voltage signal follows the output of the operational preamplifier by the ratio of $\frac{10}{25} V_O$; where 10/25 is established by the driver preamplifier input and feedback resistors. With values of resistors $\frac{RY}{RX}$ selected to equal $\frac{10}{25}$, the voltage at point A will be zero with respect to

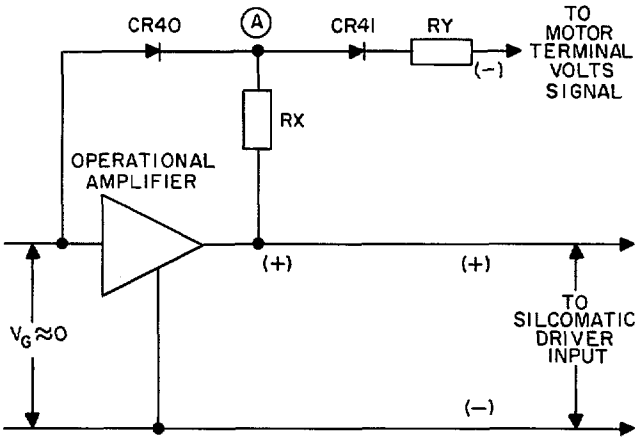


Fig. 14. Slowdown clamp circuit

common. The voltage drop across diode CR41 raises the voltage at point A slightly positive, and diode CR40 blocks.

If the Silcomatic load current goes to zero, the output of the TA decreases in attempting to slow down the drive and regain speed control. The reduction in V_O causes point A to go slightly negative and CR40 conducts. With CR40 in the conducting state, the TA is stabilized and remains about 2.5 volts below the level required to turn on the Silcomatic drive. When either the speed reference is raised or the motor speed reduced, the output of the speed preamplifier regains control of the motor speed.

METER AND TEST FEATURES

A 30-0-30 d-c voltmeter is provided at the output of the operational amplifier. This meter, visible from the front panel, indicates the preamplifier output voltage in all modes of operation.

Test points are available at the reference, operational amplifier output and auxiliary function amplifier output. These test points designated JBL1, JR2 and JBL3 respectively are mounted on the voltmeter card. Two more test points, JR4 and JR5, are mounted on the input card and monitor speed error and tach voltage. These test points can all be used while the preamplifier is operating in any of its modes. All voltage measurements are made with respect to common.

In addition to the above test points available during normal operation of the preamplifier, a test feature is included which allows the individual cards within the preamplifier to be checked on a go/no-go basis. When the preamplifier is prepared for this test mode, the drive coast-stop relay is de-energized which opens the motor contactor and prevents drive operation. See troubleshooting section.

INSTALLATION

No special installation instructions are required since the TA transistor preamplifier is normally mounted and wired at the factory. When mounted in the field, position the module for unrestricted air flow at a height convenient for access to the controls.

INSTALLATION ADJUSTMENTS

The following adjustment procedures are those performed at installation with all drive components assembled and wired with both a-c and d-c power available. For more complete information pertaining to the operation and adjustment of specific amplifier cards, refer to the appendix of this manual.

MAXIMUM SPEED ADJUSTMENT

This adjustment should be made prior to applying power to the preamplifier. It establishes the proper resistance on the regulated voltage side of the input comparison bridge located on the input card, IC3600SQIC1. The card details are shown in the Appendix.

The selection of a fixed resistor as shown in Fig. 15 is made to place the maximum speed-adjust potentiometer within proper range. To select the correct resistance, determine the tachometer voltage at maximum speed. Connect the proper jumper(s) specified in the table of tachometer volts and jumper connections shown in Table 2. This table is based on a reference voltage of 28.8 volts for maximum speed. (Assuming maximum reference voltage of 30V and draw pot set at 5.0. For 30-volt reference multiply all voltages in Table 2 by 30/28.8 = 1.042.) The wire jumper is placed directly on the input card to screw terminals labeled A through G.

TABLE 2

JUMPER CONNECTIONS ON INPUT CARD FOR CORRESPONDING TACHOMETER VOLTAGE

JUMPER	REGULATED VOLTAGE (TACHOMETER VOLTAGE)
A-D	45-62
A-C; D-G	52-70
A-C; D-E	67-85
A-C; E-G	80-98
A-C	95-113
A-B; C-G	106-124
A-B; C-E	121-139
A-B; C-D; E-G	134-152
A-B; C-D	149-167
A-B; D-G	157-175
A-B; D-E	173-191
A-B; E-G	185-203
A-B	200-218
B-D; E-G	216-234
B-D;	232-250
B-C; D-G	239-257
B-C; D-E	254-273
B-C; E-G	267-285
B-C	283-301
C-G	293-311
C-E	309-327
C-D; E-G	321-339
C-D	337-355

To establish the maximum regulated voltage at the desired level, set reference voltage at 50 percent of maximum. With the control system operating, adjust the maximum-speed potentiometer, RH20, located on the input card to the point where the

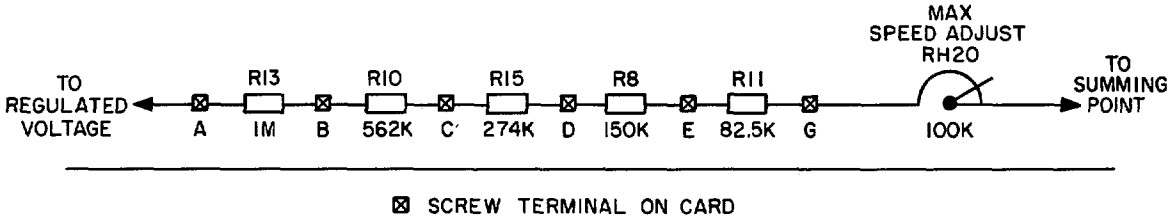


Fig. 15. Fixed resistors on input card to place maximum speed adjust potentiometer in proper range

regulated voltage is at the desired level, or where the speed of the motor is correct. Potentiometer RH20 is then locked at the desired setting.

PRECONDITION ADJUSTMENT

This adjustment should be made after the maximum-speed-adjust potentiometer is set at the desired value. Before starting the drive, turn the precondition trimpot, RH39, mounted on the auxiliary function card until the output of the operational amplifier is between +0.1 and +0.2V in the precondition mode. When operating correctly, RH39 should control the output of the preamplifier between ±0.5 volt. A clockwise rotation of RH39 produces a positive-going voltage at the preamplifier output unless CR1 or CR4 on the antihunt card clamps the output through the normally-closed ECX interlock. See the preamplifier elementary diagram, Fig. 19.

TIME CONSTANT ADJUSTMENT

The time constant produced by the product of the antihunt capacitor around the operational amplifier and the total resistance from the slider of the gain-adjust potentiometer back to the input of the preamplifier should be set to cancel the major time constant of the regulator. Before applying power to the control system, the estimated time constant should be set up on the antihunt card. Fig. 16 shows the resistor and capacitor selections available.

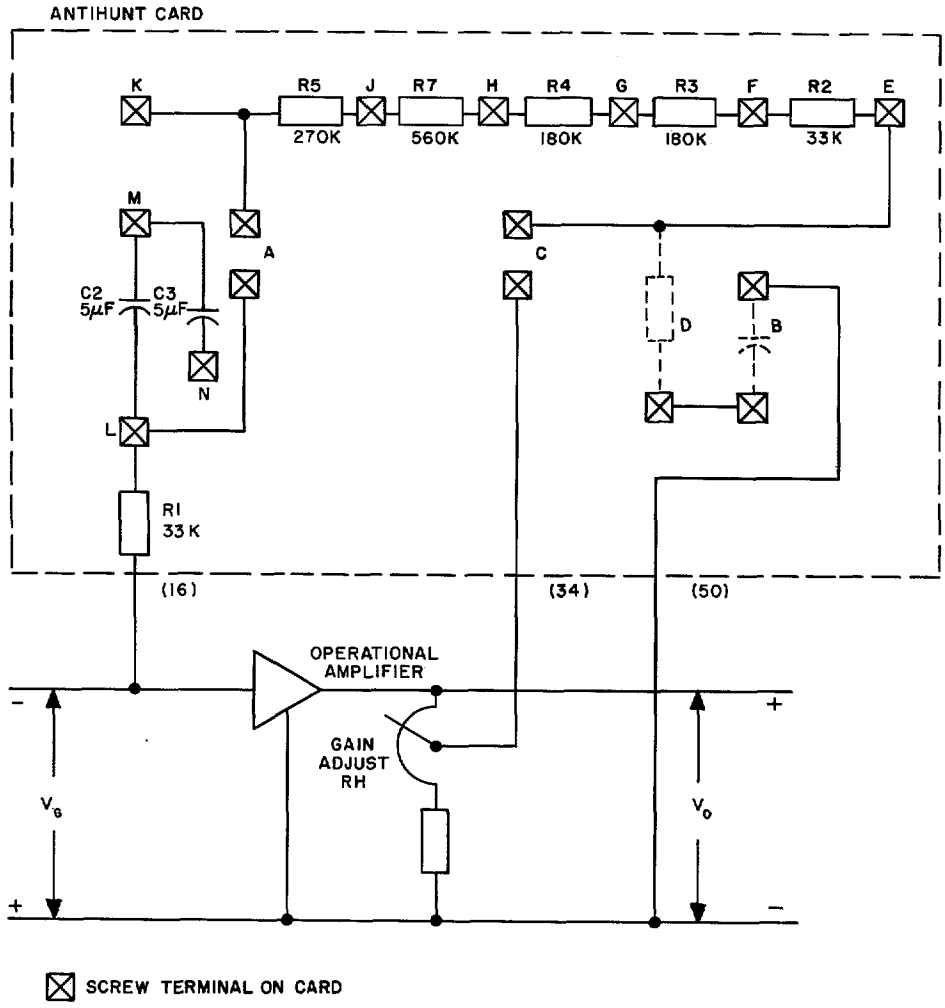


Fig. 16. Antihunt card layout for establishing proper time constant components

A jumper across position "C" is required to connect the resistor string to the gain potentiometer arm. Capacitor C2 can be connected by jumping "K" to "M" to provide 5 microfarads for an antihunt capacitance; or C2 and C3 may be connected in series or parallel to produce 2.5 to 10 microfarads respectively for the antihunt capacitance. Position "A" can be used to mount smaller capacitors if required for antihunt purposes. Also, position "C" can be used to mount additional resistors if required.

The standard configuration which stabilizes most drives is made by jumping C2 in series with the preamplifier feedback for 5 microfarads antihunt capacitance and connecting resistance in series or parallel to produce the desired time constant. Table 3 shows the value of effective resistance and corresponding jumpers required to produce the indicated time constants.

Positions "B" and "D" on the antihunt card are provided for additional stabilization. These connections are not used in most applications because of the interaction with the normal time constant except on high-response drives, or where stabilization requires these additional time constants. Refer to instructions for the specific drive.

GAIN ADJUSTMENT

The gain-adjust potentiometer, RH, is located on the voltmeter card. By varying the amount of feedback to the operational amplifier, it controls the frequency response of the regulator. This adjustment is described in the instructions for the specific drive system.

ACCELERATION ADJUSTMENT

During acceleration on the auxiliary function preamplifier, the rate of change of tachometer voltage is controlled by the magnitude of capacitance connected from the tachometer to the rate input of the auxiliary function amplifier. The accelerating capacitors C11-C14 are mounted on a terminal board secured to the rear panel of the enclosure. Each capacitor is rated 2 mfd. By connecting all four capacitors in parallel, a total capacitance of 8 mfd can be obtained. When a combination of series and/or parallel connections are used 8, 4, 2, 1 and 0.5 mfd can be obtained.

Before evaluating the actual response of the regulator in the acceleration mode, the regulator preliminary adjustment should be established for operating conditions. Table 4 gives the acceleration in volts per second for various capacitor connections.

The total acceleration time from zero to maximum speed for a given value of accelerating capacitor can be determined by dividing the maximum tachometer voltage by the volts/sec. value corresponding to the capacitor connection used. For a 200-volt tachometer at top speed, 2 mfd of acceleration capacitance will give $\frac{200 \text{ Volts}}{8 \text{ Volts/sec}}$, or 25 seconds accelerating time.

SLOW AND REVERSE ADJUSTMENT

Both slow and reverse adjustments are potentiometers located on the auxiliary function card and identified as RH40 and RH41 respectively. On non-regenerating Silcomatic drives, only the slow adjustment is set since the reverse adjustment is also established by trimpot RH40. The reverse adjustment potentiometer, RH41, is not normally used in non-regenerating Silcomatic drives. The slow adjustment sets the magnitude of the regulated voltage in the slow mode from +1 to +14 volts. This setting is independent of the input comparison bridge. The reverse voltage will be the same as the slow voltage on non-regenerating Silcomatic

TABLE 3

R/C CONNECTIONS FOR SPECIFIC TIME CONSTANTS

Antihunt Capacitance (Mfd)	Effective Resistance	Time Constant (sec.)	Jumpers Required (All Require Position C Jumper)
5	33K	0.16	M-E
5	66K	0.3	M-F
5	123K	0.6	M-G, H-F, F-E
5	213K	1	M-G, F-E
5	393K	2	M-H, F-E
5	593K	3	M-J, H-E
5	806K	4	M-J, G-F
5	1.04M	5	M-K, G-E
5	1.26M	6.3	M-K

drives. For M-G set drives and full-regenerating Silcomatic drives, the reverse adjustment is used and RH41 will set the magnitude of the regulated voltage in the reverse mode from 0.3 to 1.2 times the regulated voltage in slow.

TABLE 4

ACCELERATION CAPACITOR CONNECTIONS

Capacitor Connection	Capacitance (MFD)	Acceleration (Volts/Sec) ±20%
Four parallel	8	2
Two parallel	4	4
One capacitor	2	8
Two series capacitors	1	16
Four series capacitors	0.5	32

SLACK TAKEUP ADJUSTMENT

The slack-takeup adjustment is made by potentiometer RH21 located on the input card. When a -50V is connected to the slack takeup bridge by the operator's push button, the regulated voltage can be increased from 0 to 3 percent, depending on the setting of RH21. When +50V is connected to the bridge, a decrease in regulated voltage from 0 to 3 percent can be obtained.

On non-reversing Silcomatic drives it may be desirable to time delay the slack takeup signal to allow for finer control of the change in section motor speed. This time delay is used when the drive friction will not slow down the motor fast enough under zero current motor operation. To delay the slack takeup signal, move capacitors C42 and/or C43 on the component card in slot A1C as required from the normal location shown in Fig. 17. The capacitors are reconnected between terminals 37 to 39 (position H) and terminals 47 to 49 (position K) respectively.

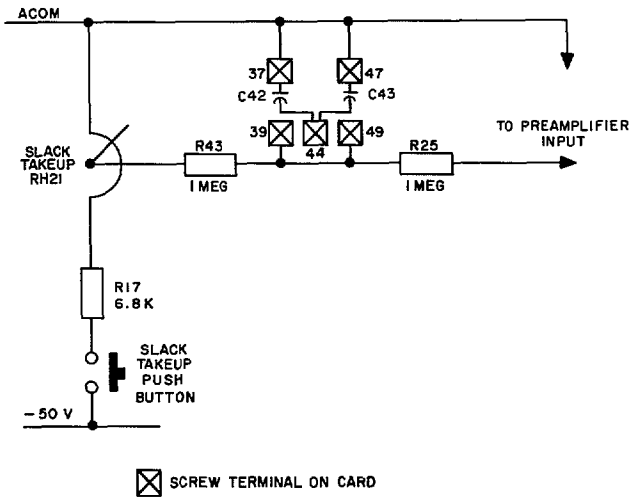


Fig. 17. Locations of slack takeup capacitors on voltmeter card in slot A1C

PREAMPLIFIER ZERO ADJUSTMENT

This adjustment is controlled by the two trimpots mounted on the operational preamplifier card. These adjustments are set at the factory and should not require field setting.

SLOWDOWN CLAMP REMOVAL

On M-G set speed regulators and full-regenerating Silcomatic speed regulators the slowdown clamp is removed. Disconnect and remove the jumper on the component card (194A3395G1) in slot A1D between points 22 and 25 (position J).

SPECIAL FEATURES

The preamplifier contains provisions for mounting additional regulated voltage feedback components. These include space for a resistor, position A, on the auxiliary function card as shown on the elementary diagram, Fig. 19. This circuit is used only in special cases.

Additional inputs are provided on the preamplifier for signals which modify speed. These input connections are R25 and R34 on the auxiliary function card and CR2, CR3 on the antihunt card. These components are used only in special cases.

SPECIAL CONNECTIONS

Coater Operation

It may be desirable to have the TA speed-regulated section accelerate to run speed using the master reference rather than the internal accelerate-to-run function. In this mode of operation, the following changes should be performed in the TA preamplifier:

NOTE: *With these connections established, the operation is from OFF to SLOW, OFF to REVERSE, or OFF to RUN.*

1. Change the metal jumper between A1M(4) to A1M(5) and relocate between A1M(5) to A1M(6).

2. Remove capacitor C2, 0.5 mfd, from the saddle clamps on the input card (IC3600SQIC1) in slot A1L.

3. Remove capacitor C41, 2 mfd, from the saddle clamps on the component card (194A3395G1) in slot A1D.

TROUBLESHOOTING

GENERAL

In the event of trouble with the TA transistorized preamplifier, the built-in test feature should be used unless the cause is obvious, due to failed components. When the test card (194A3997G1) is plugged into slot A1J after removing jumper card (194A3996G1), specific tests can be performed on the preamplifier. See Appendix E and F for elementary diagrams of jumper and test cards respectively. The circuit test points are shown in Fig. 18 and described as follows:

Test Points

P1 – Measures the positive 50-volt d-c supply.

P2 – Measures the negative 50-volt d-c supply.

P3 – By applying -50 volts to the normal reference input of the input comparison bridge, continuity to the bridge and regulated voltage source can be established. Also, continuity from the slider of the maximum-speed potentiometer can be observed by measuring the proper voltage on P3.

P4 – The operational amplifier is connected to a known input test resistor and a 33K feedback resistor mounted on the antihunt card. With the -50V bus as an input voltage, the output voltage from the amplifier can be checked on P4 to determine whether the amplifier is operating properly.

P5 – The output voltage of the operational amplifier is impressed on the gain-adjust potentiometer. This test point checks the continuity of the gain-adjust potentiometer circuit.

P6 – The output voltage of the operational amplifier is impressed on the input to the auxiliary function amplifier to determine whether the auxiliary function amplifier is operating correctly. This input voltage saturates the auxiliary function amplifier and the saturation voltage is checked on P6. When the jumper card is reinserted, the test point JBL3 should be checked with the preamplifier in the precondition mode for a complete check on the auxiliary function amplifier.

P7 – Common

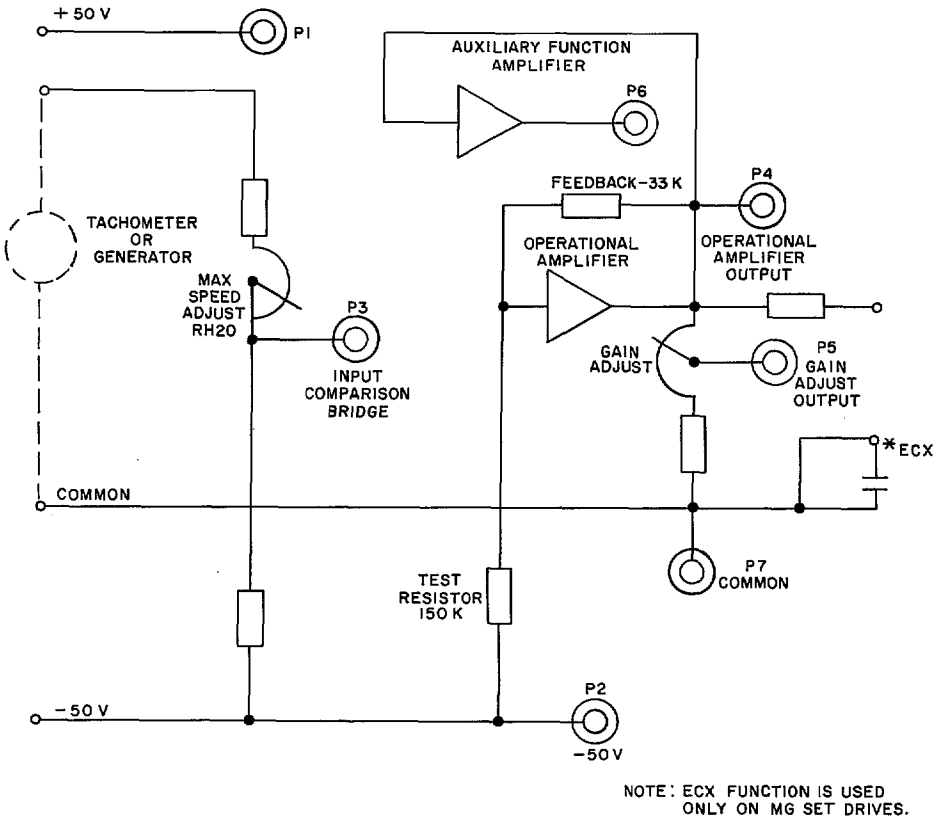


Fig. 18. Circuit test points from test card

PROCEDURE

All test voltages can be measured with a multimeter having a sensitivity of 20,000 ohms per volt, or greater. Typical voltages obtained by use of the test card are described in Table 5. Steps 1 through 4 outline the preamplifier tests using the test card while Step 5 describes an additional test performed after the test card has been removed and the jumper card reinserted in location A1J. Table 6 shows voltages obtained during normal operation of the preamplifier.

1. Shut down drive. Allow motor to come to rest.

2. Remove Jumper Card PWB-194A3396G1 from location A1J.

3. Take Test Card PWB-194A3397G1 stored in location A1A and insert in A1J.

4. With test meter, check the following points with respect to P7 on the test card according to Table 5. Take the indicated action. These tests should be made in the proper sequence since some of the test voltages are used to supply the next test.

5. Remove the test card and reinsert jumper card into position A1J. The output of the auxiliary function preamplifier at JBL3 should be between 0 and +0.2 volts. If not, replace auxiliary function card.

TABLE 5

TEST VOLTAGES OBTAINED FROM TEST CARD

Test Point	Voltage with Respect to Common (P7)	Action if Voltage not Available
P1	+50V, ±1V	1. Check external power supply. 2. Check power supply connections.
P2	-50V, ±1V	1. Check external power supply. 2. Check power supply connections.
P3	-27 to -42 Volts; depends on maximum tachometer voltage and setting of maximum speed potentiometer.	1. If zero, replace input card. 2. If greater than 42 volts, open the regulated voltage input lead; if P3 stays constant, the regulator feedback circuit is open. Check for continuity.
P4	+11 Volts, ±0.5V	1. If this voltage is not available, replace operational amplifier card. 2. If voltmeter VM on Voltmeter card does not read 11 volts, replace voltmeter card.
P5	+11 to 1V, ±0.5V as gain adjust is rotated from 0 to 10.	1. Note initial setting so as not to disturb stability of regulator. If no voltage is available, the slider of gain adjust is open; replace voltmeter card.
P6	+3 to +8.0V	1. If this voltage is -8 volts or more negative, check ECX interlock connecting output of amplifier with input to auxiliary function card. If there is continuity, replace auxiliary function card.

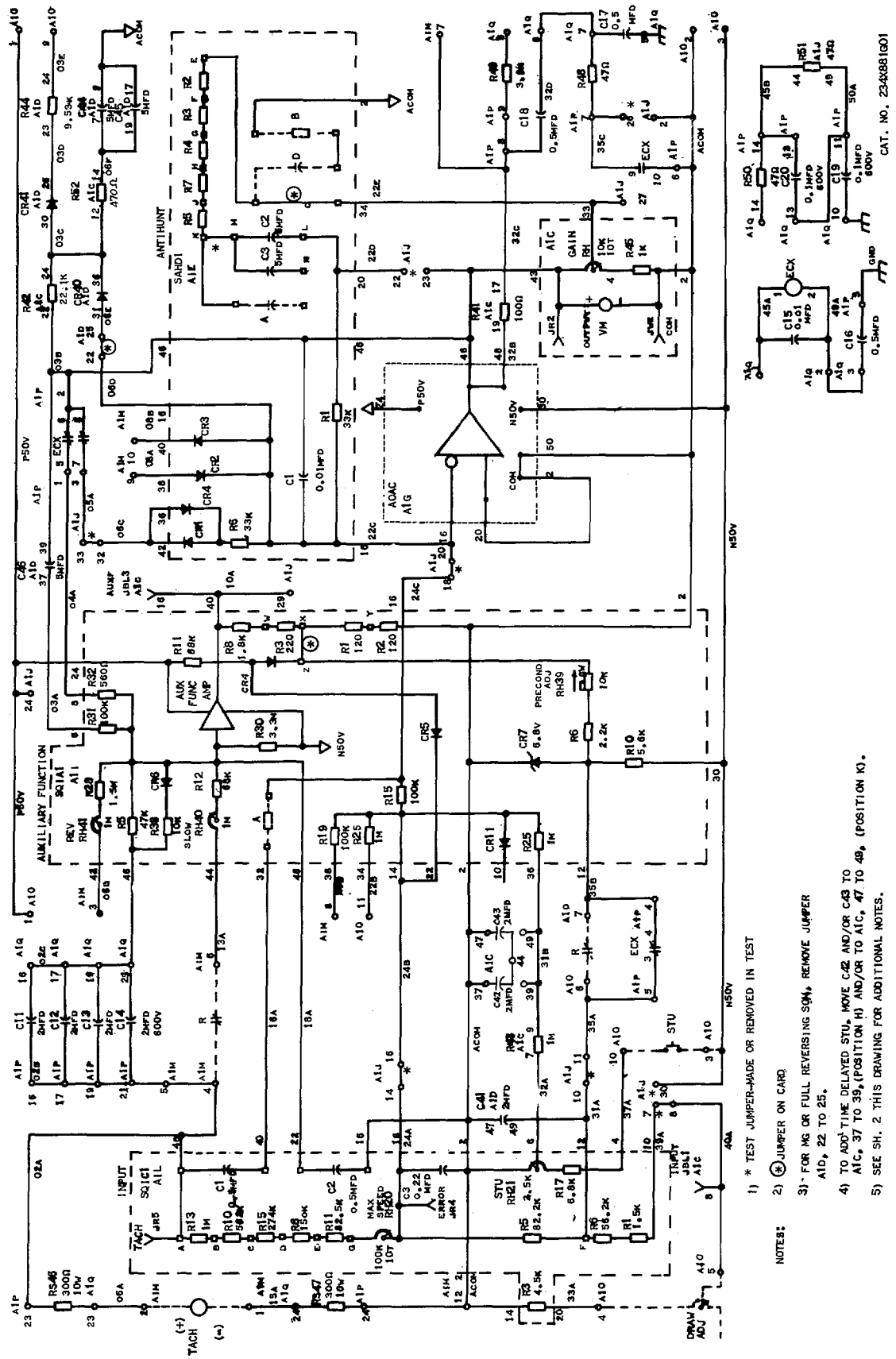


Fig. 19. TA transistor preamplifier, elementary diagram. (This diagram is typical. Always refer to elementary diagram furnished with equipment.)

TABLE 6

TYPICAL OPERATING VOLTAGES

Function	Test Point	Volts (DC)	Remarks
Positive Bus Voltage	A1O (1)	+50	
Negative Bus Voltage	A1O (3)	-50	
Input	JBL1	0 to -30	Depends on Reference Input
Output	JR2	0 to +0.5 0 to +30	In precondition mode In run mode
Auxiliary Function Output	JBL3	0 to +0.2 0 to +3.0 -8.0 or less	In precondition mode In acceleration In run mode
Operational Amplifier Input	JR4*	±0.05	For non-varying reference in run mode
Tachometer Voltage	JR5	0 to +350V	Depends on reference voltage and setting of comparison bridge.

*Use high impedance input meter (one megohm or greater)

APPENDIX A

AUXILIARY FUNCTION AMPLIFIER CARD (IC3600SQIA1)

CHARACTERISTICS

1. Power Input . . . . . 1. +50V
   2. -50V
2. Input Impedance (pin 48). . Approx. 3.3 megohm
3. Input Current . . . . . Approx. 15 microamperes
4. Output Impedance . . . . . Less than 1000 ohms
5. Gain . . . . . Set by jumper connection on card
   W-Z Connected . . . . . Approx. 0.2 volt/volt
   X-Z Connected . . . . . Approx. 0.1 volt/volt
   Y-Z Connected . . . . . Approx. 0.05 volt/volt
6. Card Size . . . . . Requires two card slots
7. Drift . . . . . Less than 1MV/^{o}C at input

APPLICATION DATA

This card contains a unity-gain buffer amplifier, a low-impedance output attenuator and a number of inputs.

The low output impedance permits signals applied at the input to "take over" control of the normal high-impedance input signals to a paper mill preamplifier. This "take over" function is through a diode clamp circuit so the normal inputs are not disturbed when the input to the auxiliary function amplifier is negative.

When used to control acceleration of a section, an external capacitor is connected from the section tachometer to pin 46. The rate of change of tachometer voltage will be approximately $\frac{15}{c}$ volts/sec. (C is the feedback capacitance in microfarads.)

Adjustable inputs for "slow," "reverse," and "precondition" are provided on the card.

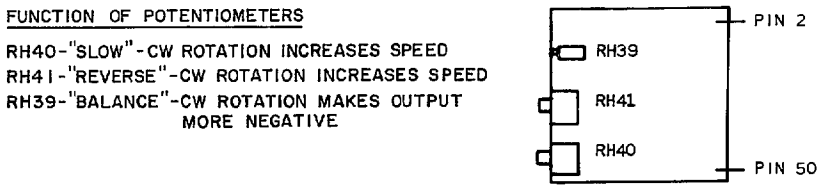


Fig. A-1. Location of potentiometers on auxiliary function amplifier card.

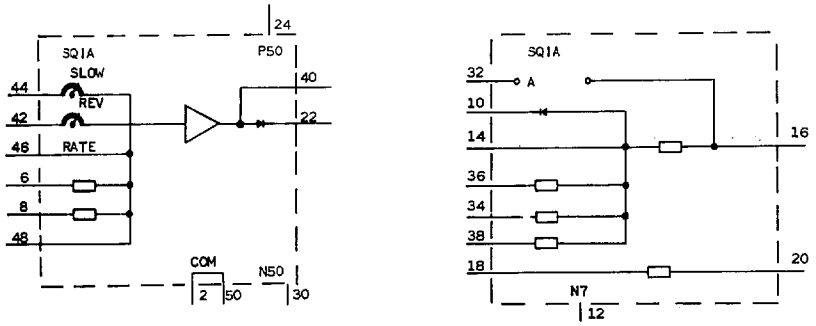


Fig. A-2. Pin connections on auxiliary function amplifier card

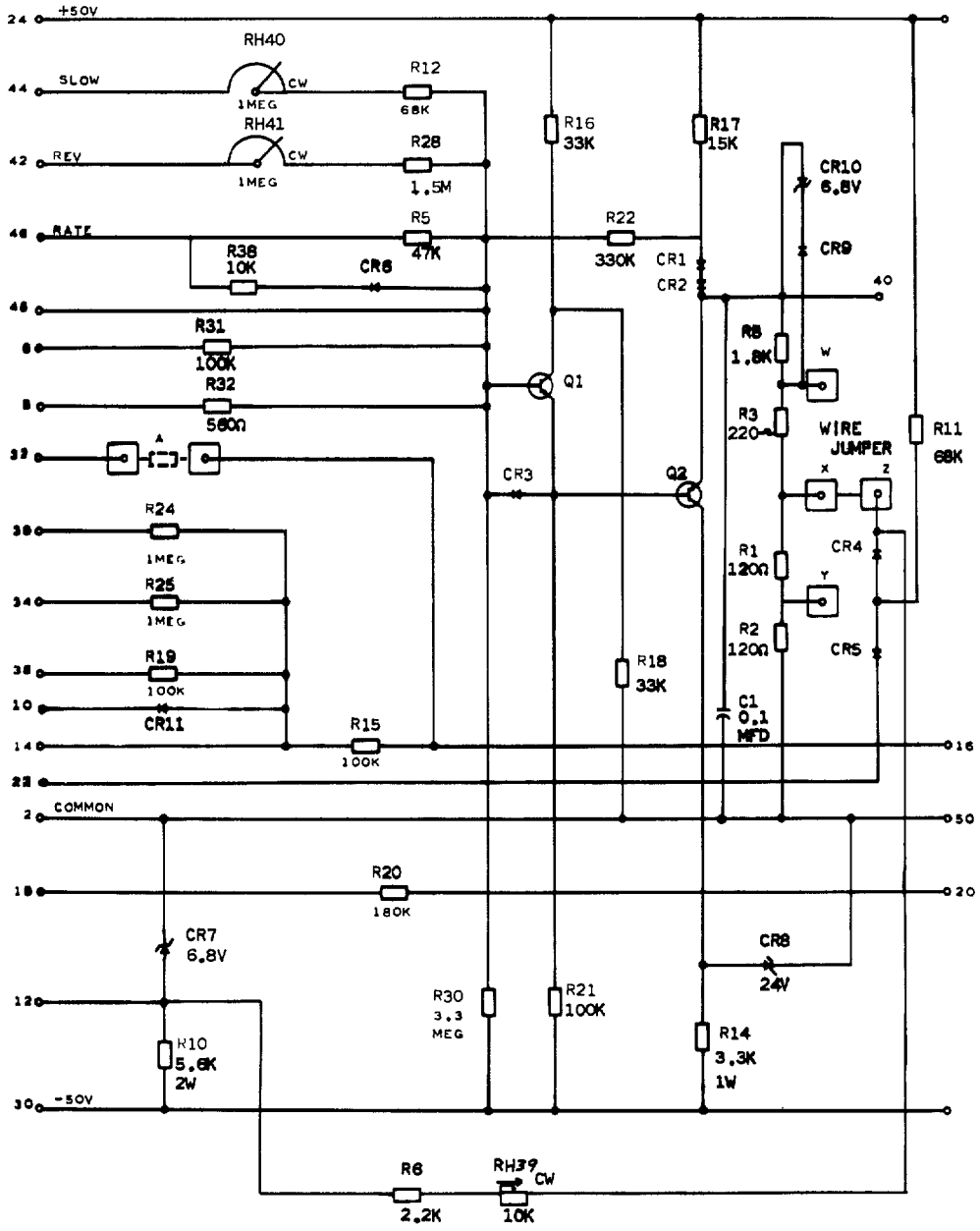


Fig. A-3. Auxiliary function amplifier elementary diagram

APPENDIX B
OPERATIONAL AMPLIFIER CARD
(IC3600AOAC1)

AMPLIFIER CHARACTERISTICS

1. Power Input	1. +50V, 24MA + load current 2. -50V, 30MA + load current		
2. Signal Input	1. $\pm$.66MV for full output range (DC) 2. $\pm$ 6.6 x 10^{-9} amps for full output range (DC)		
3. Output	$\pm$30V, $\pm$10MA (see application data)		
4. Frequency Response			
A. 10K input and feedback	285KC typical		
B. 100K input and feedback	65KC typical		
C. 1MEG input and feedback	7.6KC typical		
D. Gain Bandwidth (Open Loop)	Greater than 800KC		
5. Temperature Drift	Input Current		Input Voltage
A. 0°C to 25°C	-1.5 $\pm$2 x 10^{-9} amps/°C		$\pm$20 x 10^{-6} volts/°C
B. 25°C to 70°C	$\pm$2 x 10^{-9} amps/°C		$\pm$20 x 10^{-6} volts/°C
C. 24 hour drift (25°C)	$\pm$25 x 10^{-9} amps		$\pm$250 x 10^{-6} volts
6. Recovery from 100V Input	Recovers to 1 percent of final value in		
A. 10K input and feedback	.06 millisec. (typical)		
B. 100K input and feedback	.10 millisec. (typical)		
C. 1MEG input and feedback	.40 millisec. (typical)		
7. Recovery from Shorted Output	Recovers to 1 percent of final value in		
A. 10K input and feedback	.2 millsec. (typical)		
B. 100K input and feedback	.15 millisec. (typical)		
C. 1MEG input and feedback	.4 millisec. (typical)		
8. Output Impedance (Open Loop)	Typically less than 10 ohms from DC to beyond 1 megacycle		
9. Maximum Capacitive Load	680 PFD		
10. Frequency Limit at Maximum Amplitude			
A. Resistive load (3.3K)	70 KC (typical) (12V/microsec. Slew rate)		
B. 680 PFD load	35 KC (typical) (6V/microsec. Slew rate)		
C. 100 ohm, .01 MFD series load	3.5 KC (typical)		
11. Noise at Output (100K input and feedback)	3 millivolts due to amplifier alone.		
12. Common Mode Input	$\pm$8 volts maximum		
13. Common Mode Rejection	Can be adjusted for more than 120 DB at DC. See application notes for circuits to be used.		

APPLICATION DATA

Theory of Operation

The operational amplifier is a high-gain amplifier with a controlled "single break" frequency response. An inverting input and a non-inverting input are provided. Normally, the non-inverting input is tied to common of the analog power supply. A feedback resistor, capacitor, or series RC circuit, from output to inverting input will provide a stable closed loop which will respond in a controlled manner to any signal or signals applied to the inverting input.

Because of the high amplifier gain, the amplifier output voltage will be very nearly equal to the negative of the input voltage times the ratio of the feedback impedance to the input impedance. If the input, or feedback circuits, are networks the inverting input can be considered as being at analog common potential. The sum of the currents from input voltage to inverting input and from output voltage to inverting input can be equated to zero and the equation solved for $\frac{V_{out}}{V_{in}}$.

Effect of Temperature Variation

The amplifier is adjusted in test to have zero input voltage and current error at the temperature prevailing when the adjustment is made. This will normally be done at 25 to 35°C. When the amplifier is operated at a different ambient temperature, there will be an error voltage and current at the amplifier input because the temperature compensation circuits do not exactly compensate for the transistor characteristics. The amount of error cannot be predicted accurately unless each amplifier is pretested through the entire temperature range. The drift data given on the characteristics sheet are based on test performed on four prototype amplifiers.

The calculation shown below illustrates how the characteristic data would be used to calculate the offset band within which all amplifiers are expected to operate. Fig. B-3 shows the offset band, the calculated drift of four amplifiers based on input drifts measured in a previous test, and measured offset with this circuit.

Temperature Drift of AOAC Preamplifier

1. The amplifier input and feedback circuit with drift elements is shown in Fig. B-1.

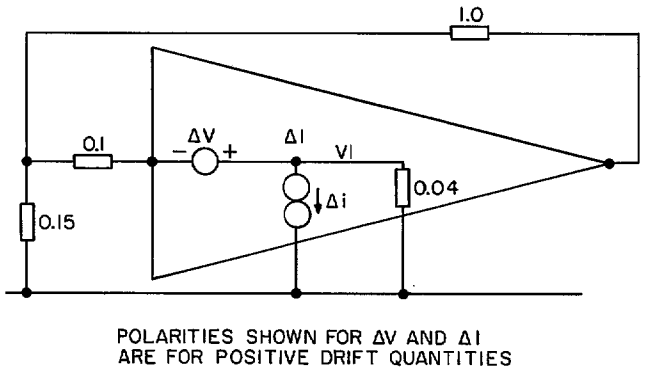


Fig. B-1. Operational amplifier input and feedback current

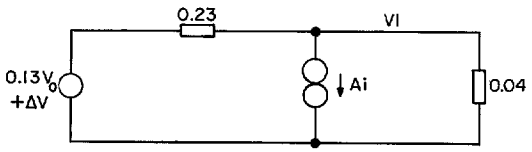


Fig. B-2. Operational amplifier equivalent circuit

2. The equivalent circuit is shown in Fig. B-2.

3. Equations relating V_o, ΔV, and Δi for V_1 = zero.

$$\frac{.13\ V_o + \Delta V}{.23} = \Delta i$$

$$V_o = \frac{.23\Delta i - \Delta V}{.13} = 1.77\,\Delta i - 7.7\,\Delta V$$

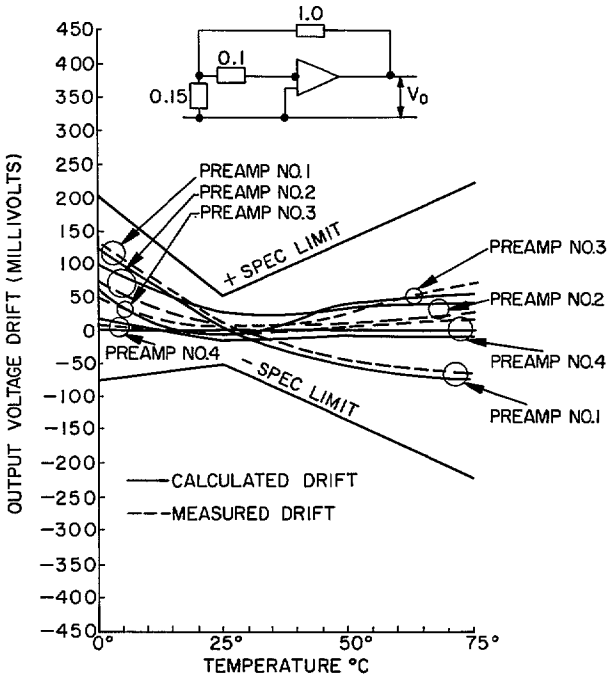


Fig. B-3. Temperature test characteristics of four prototype operational preamplifiers

If Δi is in nano-amperes and ΔV in millivolts, then V_0 will be in millivolts.

If the amplifier is expected to operate over a temperature range of 25°C to 50°C, the specification sheet gives $\Delta i = \pm 25 \times 10^{-9} \pm 25 \times 2 \times 10^{-9}$ or ± 75 nano-amperes; similarly, $\Delta V = \pm 25 \times 20 \times 10^{-6}$ $\pm 250 \times 10^{-6} = \pm 750 \times 10^{-6}$ or $\pm .75$ millivolts. These values substituted in the above equation would give the maximum offset. The tracking error for two drives would be twice this value.

Noise Protection

The amplifier has an input filter, a low output impedance, bus filters and a separate Zener regulated and filtered bus system for the input stages of the amplifier. All of these features help to reduce the susceptability of the amplifier to noise transients. Large noise inputs can still cause the amplifier to saturate. To protect the amplifier from these large noise inputs it is essential that the analog bus system (especially analog common) must be stable and maintained very close to earth or enclosure potential. A direct tie from bus to enclosure should be made at one point only but distributed capacity from the common bus to enclosure will greatly reduce the effect of high frequency transients.

If a stable bus system is maintained, filters can be added which will reduce the effect of signal transients as well as high frequency ground loop currents. Some recommended filters are shown below.

1. When the amplifier output must be wired to a remote location (Fig. B-4) where it may be subjected to noise, it is recommended that a series 100 ohm resistor be mounted on the page, near the amplifier. A suitable capacitor filter should be mounted where the output leaves the controlled low noise area of the panel. The output should be carried through a twisted shielded pair to the load. It will frequently help to ground both ends of the shield if the OUTPUT LINES ARE SUBJECTED TO HIGH FREQUENCY NOISE.

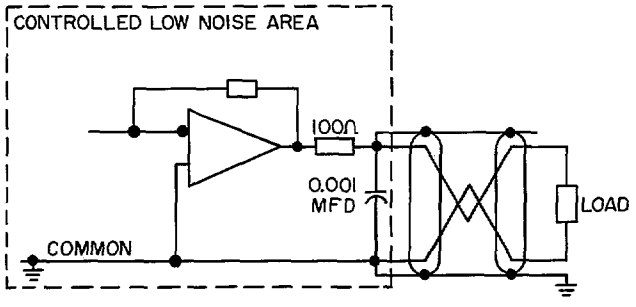


Fig. B-4. Noise filter and output shielded pair

2. Protection for input signal wires can be provided in a similar manner. It is recommended that 10 to 20 percent of the input resistance be added in series with the input line where it enters the controlled low noise area. Also a suitable filter capacitor should be mounted close to this resistor. Fig. B-5 illustrates this filtering.

This filter can be used to reduce high frequency transients. It will not be satisfactory for noise frequencies which are equal or lower than the signal frequencies to be controlled.

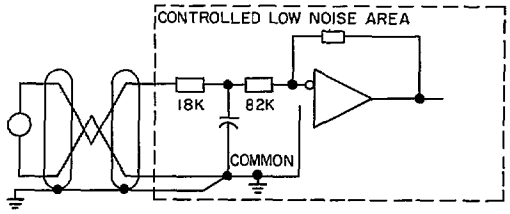


Fig. B-5. Noise filter and input shielded pair

Common Mode Rejection

DC or low-frequency common mode signals can be appreciably attenuated with a circuit as shown in Fig. B-6.

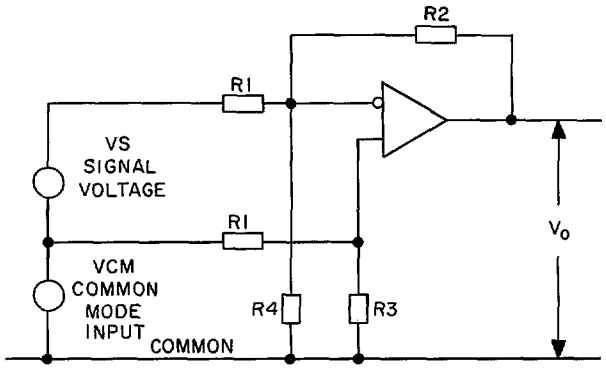


Fig. B-6. Circuit for attenuating low-frequency common-mode signals

For this configuration $V_0 = -V_s \frac{R_2}{R_1}$

Also Vcm (max.) x $\frac{R_3}{R_1 + R_3} \leq 8$ volts.

R_4 is approximately $\frac{R_3 R_2}{R_3 + R_2}$. It is recommended that R_4 should consist of a fixed resistor and a series rheostat so that a ± 10 percent adjustment can be made. For DC common mode signals the common mode rejection can be set as high as 120 DB. At 60 cycles, the rejection ratio is reduced

because of the amplifier stabilizing is not balanced for both inputs. An additional RC circuit in the inverting input can be used to improve the high frequency common mode rejection. This requires careful selection of components and is not recommended if a high degree of AC common mode rejection is required. This circuit shown in Fig. B-7 can be easily adjusted to give up to 72 DB rejection of 60-cycle common mode voltage.

For this configuration $V_O = -V_S \frac{R_3}{R_1}$

Also Vcm (max.) x $\frac{R_2}{R_1} \leq 30$ volts

The rejection ratio is independent of impedance in series with the common mode input. Impedance of the signal source may reduce the common mode rejection. The resistor R_1 identified by an asterisk above should be adjustable to compensate for signal source impedance.

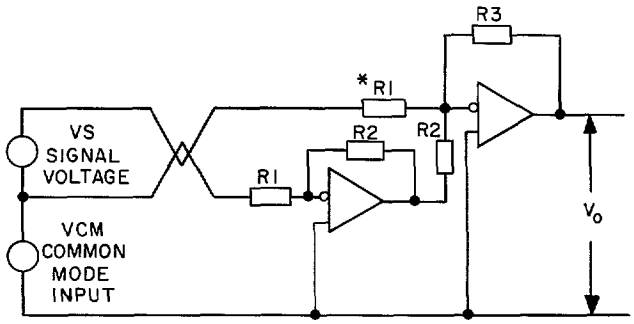


Fig. B-7. Circuit for attenuating high-frequency common-mode signals

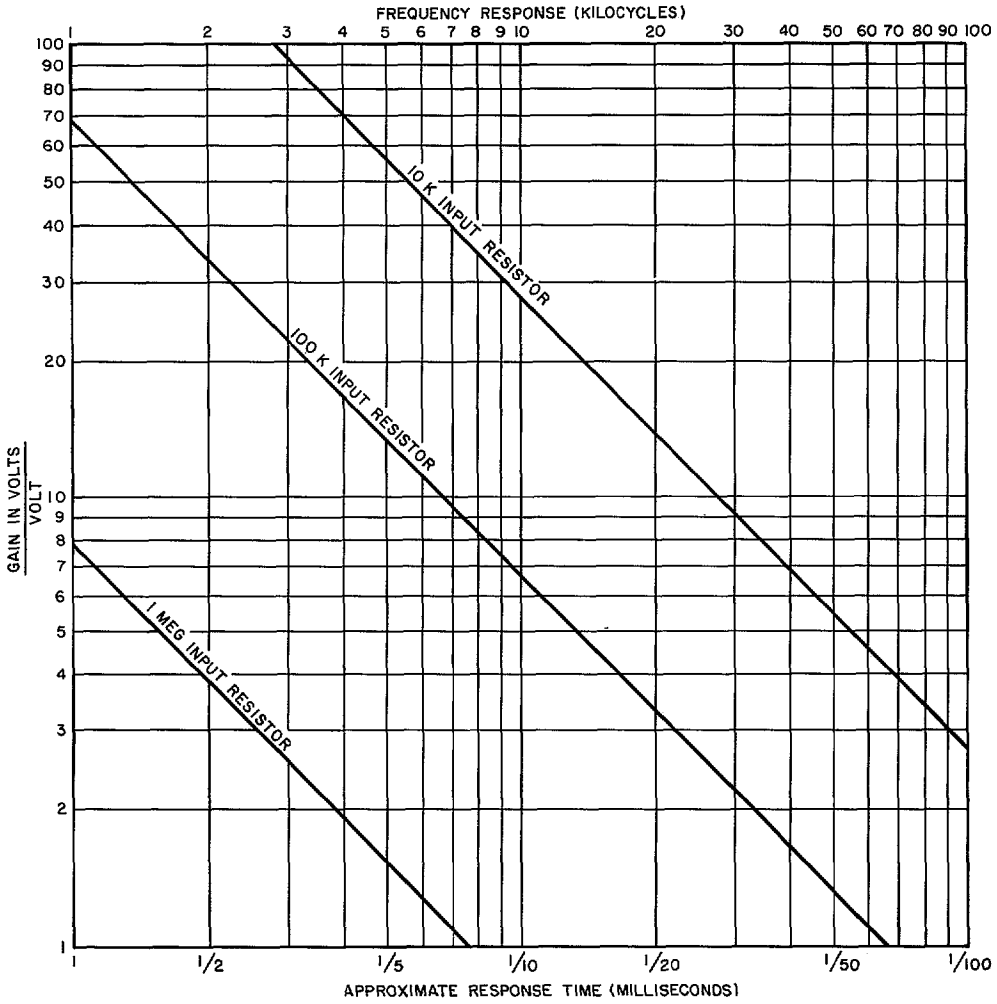


Fig. B-8. Operational amplifier typical frequency response curves. Graph also shows response time for 99 percent recovery from a step input. This information is subject to considerable error above 100 kc because of a stray capacitance of wiring and capacitance of resistors in the input and feedback circuits.

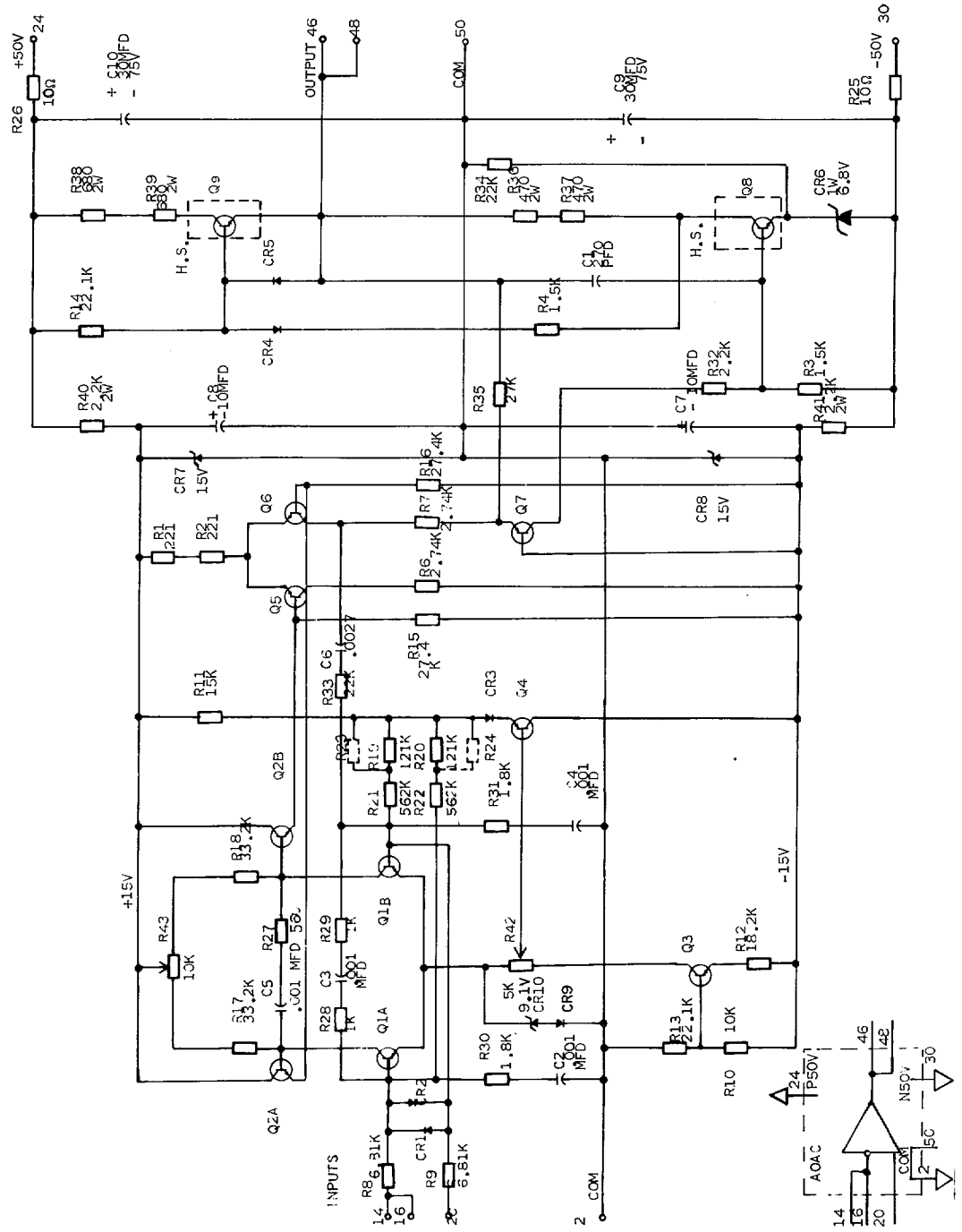


Fig. B-9. Operational amplifier (IC3600AOAC1), elementary diagram

APPENDIX C
ANTIHUNT CARD
(IC3600SAHD1)

APPLICATION NOTES

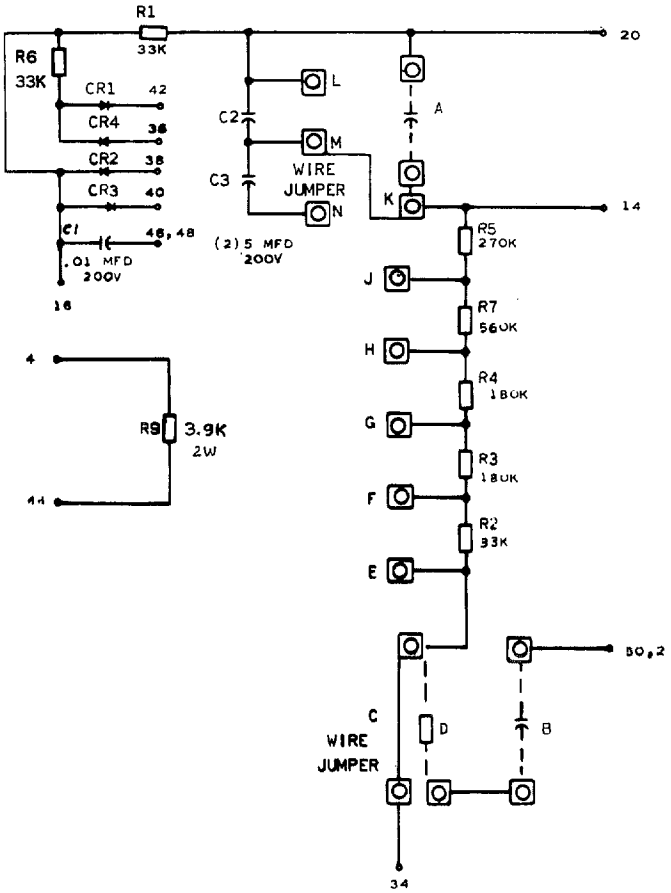
1. The antihunt circuit (Fig. C-1) is made up of resistors and capacitors which provide a means to vary the time constant normally used in making up antihunt or stabilizing networks used in sectional drive speed regulators.

2. Saddle points are available to allow adding resistance, shorting out resistance, and adding capacitance in series.

3. Saddle points are also available to allow for another time constant if necessary.

4. The 0.01 mfd capacitor C1 is used for noise suppression.

5. Diodes CR1, CR2, CR3, and CR4 are used for clamping or current limit control.



1. ALL RESISTORS ARE 5% CARBON 1/2 WATT UNLESS DESIGNATED OTHERWISE
2. C2 AND C3 ARE METALLIZED MYLAR ± 20% CAPACITY TOLERANCE. INSULATION RESISTANCE IS 30,000 MEGOHM X MICROFARADS.

Fig. C-1. Antihunt card, elementary diagram

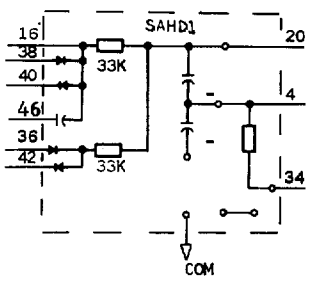


Fig. C-2. Antihunt card, pin connections

APPENDIX D
INPUT COMPARISON CARD (IC3600SQIC1)

APPLICATION NOTES

1. This card (Fig. D-3) requires two card spaces because of size of components used.

2. The input comparison circuit (Fig. D-1) is used to provide a variety of input impedances for different voltages used in a speed regulator for sectional drive motors.

V_A is normally applied to a DC amplifier in a closed-loop configuration to keep the voltage V_{A-B} essentially zero. This is done by feedback, changing V_1 to keep the ratio of $\frac{V_1}{V_2} = \frac{R_1}{R_2}$. The maximum speed potentiometer R20 is adjusted to give this precise relationship for the reference voltage and input desired.

3. Saddle clamps are available to place a capacitor in the circuit for feedback purposes. This capacitor can be up to 0.5 mfd., 600V in size.

4. The input resistors are capable of being jumped.

5. Potentiometer RH21 is used for slack takeup on the sectional drive. This is accomplished by applying an additional reference input into the amplifier.

6. Capacitor C2 is used to filter the auxiliary function amplifier input when operating from stop to run.

7. Resistor R3 is an auxiliary resistor used in conjunction with a draw potentiometer.

8. Potentiometer locations are shown in Fig. D-2.

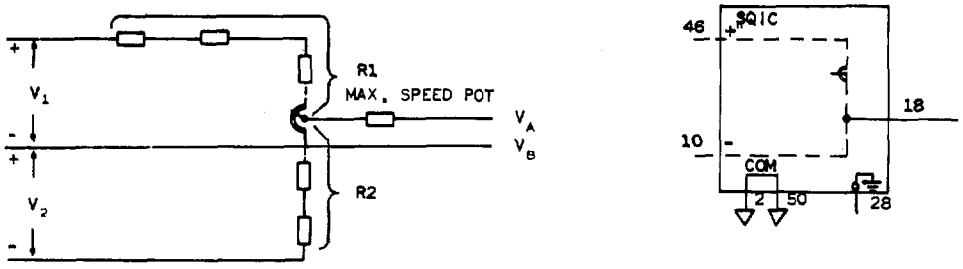


Fig. D-1. Input comparison circuit

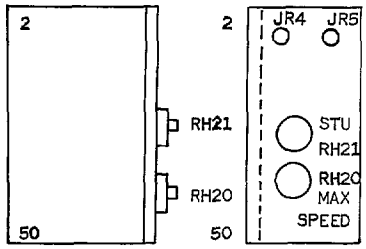
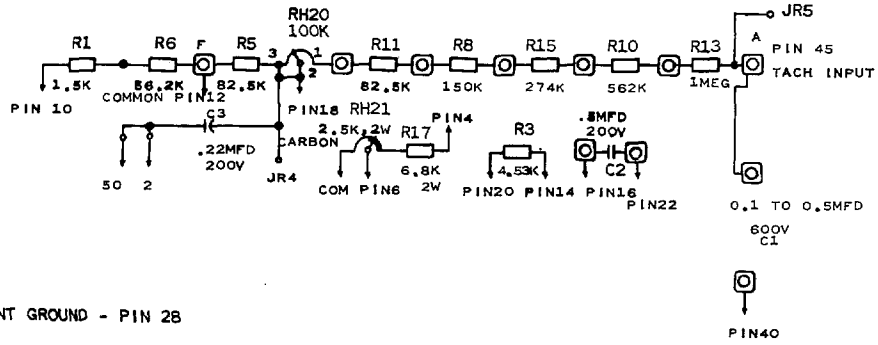


Fig. D-2. Potentiometer locations on input comparison card

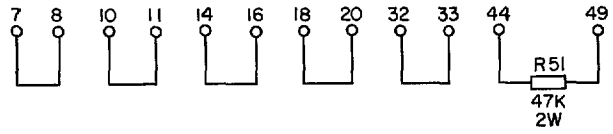


ALL RESISTORS ARE 1/2W METAL FILM, 25 PPM TEMP. CO-EFFICIENT
C2 IS METALLIZED MYLAR ±20%/

Fig. D-3. Input comparison card, elementary diagram

APPENDIX E
JUMPER CARD
(194A3396G1)

The elementary diagram of the jumper card mounted in slot A1J is shown in Fig. E-1.

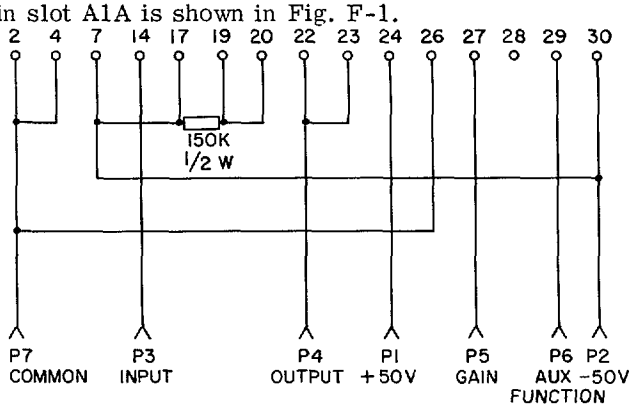


NOTE:
THE ABOVE JUMPERS ARE LOCATED ON THE JUMPER CARD. INSERTION OF THIS CARD INTO SLOT A1J SETS UP THE TA PREAMPLIFIER CIRCUITS FOR NORMAL OPERATION.

Fig. E-1. Jumper card, elementary diagram

APPENDIX F
TEST CARD
(194A3397G1)

The elementary diagram of the test card STORED in slot A1A is shown in Fig. F-1.



NOTE: THESE CIRCUITS AND TEST POINTS ARE LOCATED ON THE TEST CARD. WHEN THIS CARD IS INSERTED INTO SLOT A1J AFTER REMOVING JUMPER CARD, THE PREAMPLIFIER IS PREPARED FOR TESTING.

Fig. F-1. Test card, elementary diagram

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